

# POKHARA UNIVERSITY

Level: Bachelor  
Semester: Fall  
Programme: BE  
Course: Digital Signal Analysis and Processing

Year : 2022  
Full Marks: 100  
Pass Marks: 45  
Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Attempt all the questions.*

- a) Consider the analog signal  $x_a(t) = 3\cos(100\pi t)$  8
- i) Determine the minimum sampling rate required to avoid aliasing.
  - ii) What is the Nyquist rate for the signal  $X_a(t)$ ?
  - iii) If the signal is sampled at the rate  $F_s = 200\text{Hz}$ . What is the discrete-time signal obtain after sampling?
  - iv) What is the frequency  $0 < F < F_s/2$  of a sinusoid that yields samples identical to those obtained for  $F_s = 200\text{Hz}$
- b) Define time-invariance and causality of a discrete time system. Examine whether following systems are stable or not: 7 2
- i)  $y[n] = x[n] + 1$
  - ii)  $y[n] - y[n-1] = x[n] + x[n-1]$
- a) Find the convolution between two signals  $x[n] = a^n$ , for  $0 \leq n \leq 6$  and  $h[n] = 1$ , for  $0 \leq n \leq 4$  8 2
- b) Define Z transform. Find the inverse z transform of  $X(z) = \log(1 + az^{-1})$  7 3
- a) Why do we need DFT when we have DTFT? Determine the circular convolution of the sequence 8 4
- $x_1(n) = \{1, 2, 3, 4\}$  and  $x_2(n) = \{5, 6, 7, 8\}$
- Also verify your answer.
- b) Define Fast Fourier Transform (FFT)? Determine 8-point DFT of the sequence using Decimation in Time FFT. 7 4.5

$$x(n) = \cos\left(\frac{n\pi}{4}\right)$$

- 5 4. a) Obtain the parallel form realization of following IIR filter

$$H(z) = \frac{1 + \frac{1}{4}z^{-1}}{(1 + \frac{1}{2}z^{-1})(1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2})}$$

- 5 b) Convert the following IIR filter into lattice ladder structure

$$H(z) = \frac{1 + z^{-1} + 2z^{-2} + z^{-3}}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}}$$

5. a) Convert the following in Direct form- II realization described by

$$H[z] = (1 + z^{-1} + 2z^{-2} + 3z^{-3}) / (1 + 15z^{-1} + 15z^{-2} + 12z^{-3})$$

- 6 b) With the low pass specification  $\Omega_p = 3.2$  KHz and  $\Omega_s = 4.8$  KHz and sampling frequency  $\Omega_{fs} = 12$  KHz and  $\alpha_s = 40$  dB, find the length and value of  $\beta$  for Kaiser Window.

- 7.2 6. a) Derive Impulse Invariant Method for IIR filter design. Also illustrate the mapping from the s-plane to the z-plane while using IIM.

- 7 b) Design a Chebyshev analog filter with maximum passband attenuation of 2.5 dB at  $\Omega_p = 20$  rad/sec and stopband attenuation of 30 dB at  $\Omega_s = 50$  rad/sec

7. Write short notes on: (Any two)

a) Gibbs Phenomena 6.1

b) Basis Elements of DSP \

2 c) BIBO Stable